Digital Audio-Video Signal Processing

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1 Question - 8 pts

Draw the block diagram of a DPCM codec, explaining the need for the feedback loop. Consider a realization of a continuous valued source with memory:

$$\{10.3,\ 9.2,\ 10.8,\ 1.9,\ 2.8\ldots\} \tag{1}$$

Compute the first five samples of

- the quantization indexes of the prediction error,
- the reconstruction levels of the prediction error,
- the reconstructed sequence at the decoder,

when a midrise uniform scalar quantizer is used with quantization step size equal to $\Delta = 2$ (ignore saturation of the quantizer).
Describe viable improvements to the conventional DPCM scheme addressing nonstationarity of the input signal.

2 Question - 4 pts

Formulate the optimal rate allocation problem in transform coding, specifying the input and output data, the objective function, the constraints and the intuitive rationale behind the optimal solution.

3 Question - 4 pts

Sketch the block diagram of a perceptual audio codec (encoder and decoder), briefly illustrating the functionality of the individual blocks.

4 Question - 4 pts

Briefly illustrate motion estimation by hierarchical block matching.

5 Question - 12 pts

Load the Matlab *rice* image (hint: *I=imread(rice.png);*)
Consider the pixel values as symbols emitted by a discrete source with alphabet $\{0, 1, \ldots, 255\}$.

- Assuming that the source is memoryless, compute the empirical entropy $H(x)$.
- In order to exploit the source memory, compute the prediction error by taking the difference $z_i = x_i - x_{i-1}$, where i denotes the current pixel and $i-1$ the top neighbor (Hint: use the $a = A(:)$ operator to raster scan a 2D matrix columnwise). Compute the empirical entropy of the prediction error $H(z)$.

Now consider a generic set of natural images. For each of the following statements, say if it is true or false, justifying your answer

1. $H(x) \leq 8$
2. $E[H(z)] \leq E[H(x)]$, where the expectation is taken over a set of natural images.
3. It is not possible to have an image for which $H(z) > H(x)$.
4. $E[H(x_i|x_{i-1})] = E[H(z)]$.